



Search Personalization at Netflix

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ABSTRACT

At Netflix, personalization plays a key role in several aspects of our user experience, from ranking titles to constructing an optimal Homepage. Although personalization is a well established research field, its application to search presents unique problems and opportunities. In this paper, we describe the evolution of Search personalization at Netflix, its unique challenges, and provide a high level overview of relevant solutions.

CCS CONCEPTS

• **Information systems** → **Personalization; Recommender systems; Learning to rank.**

KEYWORDS

Recommender Systems, Search, Neural Networks, Personalization

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1 INTRODUCTION

Modern streaming services use recommender systems to help users discover and enjoy content from their large catalogs. However, in many cases users either don't find what they were looking for or have some clear intent in mind that they want to express. In such scenarios, users resort to Search to find something to play. The search engine's goal is to help them discover something delightful, satisfying their needs as quickly and easily as possible. On Netflix, streaming from search results accounts for 20% of discovery [6], and Search plays a key role in helping users discover the best title to watch.

Historically, Search and Recommendations have been treated as two separate problems; Search being mainly focused on the

query. Recommendations on the user. Personalized Search brings them together because both the query and user information can be taken into account to effectively respond to the user's needs. Search particularly can benefit from personalization as in many cases queries are broad enough to require different results for different users.

The problem of Personalized Search has been widely studied in academia [2, 3, 7], and there is also evidence of its successful application in industry. Particularly, Airbnb trained listing and hosts embeddings to personalize the listing ranking [4]. LinkedIn applied personalization to expertise search and federated search [1, 5].

In the remainder of this paper, we first introduce the Netflix Instant Search problem and its unique challenges. We then discuss different strategies for search ranking personalization.

2 INSTANT SEARCH PERSONALIZATION IN THE ENTERTAINMENT DOMAIN

In a long form video entertainment platform such as Netflix, the majority of the usage happens on TV devices. As such, it is of paramount importance to provide an effortless and enjoyable discovery experience, requiring the least amount of effort from our users. In order to reduce typing and provide a frictionless experience, Netflix adopted an "Instant Search" paradigm. That is, a full Search is performed for each keystroke. While this paradigm offers clear benefits, it has strict latency requirements which can be in part mitigated using well-tuned results caching at several layers. However, results caching is not useful when serving fully personalized results since different users will have different query results and the cache hit rate would be extremely low. We will elaborate more on how we tackled this challenge in the next section.

A unique characteristic of Instant Search on TV is that queries are very short. This presents a great opportunity for personalization as rich knowledge about the user can complement the limited query context. Users' historical preferences can help the system to better predict users' intent and in turn return a unique set of tailored results.

In [6], the authors identified two main Netflix search use cases: **fetch** and **explore**. The **fetch** use case is most common, where users have a clear intent to search for a specific title ("*I know what I want, I need you to get it for me*"). Personalization can help users get to their results faster (less typing) if the title they are looking for has high affinity with their taste profile, which is often the case. A subset of this intent is for **out-of-catalog** videos [6], where the title the user is looking for is not available. This is purely a recommendation

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use case and the system should return titles which are related to the unavailable one. Also in this scenario personalization may provide additional value to the recommended results as the set of related titles can be broad and the notion of related titles can be different for different users. The **explore** use case (*"I don't know what I want, let's understand what you have"*) consists of broad queries, such as genres. Given the broad nature of the results, personalization can help optimize for the correct relevant titles for each user.

One important aspect to highlight is that different search intents have different relevance personalization trade-offs. For example, query relevance is of primary importance for the fetch intent, since too much personalization could hurt the user experience if high affinity but lexically irrelevant results are ranked high in the list. On the other hand, for a genre query, titles belonging to the requested genre with high user affinity but low lexical query similarity are preferable. In other words, there is a delicate balance between relevance and personalization which is also intent specific.

3 RANKING AND MODELING APPROACH

The personalization approach we adopted is mainly based on augmenting our ranker features set with different variations of personalized features. In the following we describe the underlying base ranker and then the different personalization strategies we adopted considering different engineering and modeling trade-offs.

3.1 Ranker

Fig 1 shows a high level architectural view of our ranking model, which is based on a deep learning feed-forward neural network model. At a high level, features can be classified in three types: *context* features, *title* features and *context-title* features. We use the term context to denote user inputs and request specific information. Some examples of context features include the query, country and language/locale. Title features instead describe solely the target title to rank and some of the features include metadata, popularity, etc. Lastly, the context-title features include attributes describing the match between some of the contextual information and the target title, for example query-title lexical similarity is one. The features listed in Fig 1 represent the newly introduced personalized features which we will describe in the next sections. The base ranker however does not take into account any personalized features.

As briefly mentioned in the previous section, a very important aspect to consider when it comes to Search is latency, which typically corresponds to how long it takes for a query to be processed and return results. Given the strict latency requirements, our ranker relies on different caching layers for the features and results list. Particularly, feature outputs are stored in a feature cache so that we don't have to recompute the same features at each request. Results are stored in a separate dedicated cache and use the context as key, therefore all users submitting the same query will receive the same cached results. This makes unpersonalized search very efficient and low latency.

With the introduction of personalized results, we had to depart from our initial results caching scheme. In the following, we will describe the steps we have taken towards caching and the different features introduced together with modeling changes. This iterative

approach allowed us to test the impact of each individual stage in terms of both user experience and system complexity.

3.2 Cluster based personalization

We decided to take a gradual approach first by adding additional personalized context-only features. The main advantage of this approach is that we can still leverage the results caching scheme, as users with the same contextual information will receive the same cached results. This is possible because we are only changing the context which is also the key for the results cache.

In this approach, we cluster users into a small number of clusters based on their viewing history and use their cluster assignment as a personalized context feature. We then added different feature interactions terms between the cluster identifier and title content attributes. This is a coarse level of personalization (at a cohort level), since users in the same cluster will receive the same results. On one hand this is limited in terms of personalization capability, but on the other hand it is appealing from a caching perspective since we can still cache the results lists, increasing the original cache size by the number of clusters.

3.3 Personalized Affinity Feature

In this approach, we use a separate model to compute personalized user-item affinity scores, which can be interpreted as the probability the user would engage with the item. This model can be considered as a plug and play component, where we can leverage different models, which is particularly appealing from a systems perspective.

The personalized affinity score becomes a context-title feature and provides full personalization capability to the model. That is, different users will receive different results and because of that, we cannot use the results cache anymore since we would need to store each unique results set. This would require a very large cache storage and also the cache hit rate would also be extremely low making results caching pointless. Fortunately, we can still rely on the feature caching, and perform a forward pass at each request. Note that different features have different caching requirements and unpersonalized features have much lower caching key cardinality requirements than personalized data.

3.4 End-to-end model

In the previous case, we rely on a pre-trained model to provide user-item affinity. The model has a different objective than the one used for search ranking. While that setup is simpler, it does not ensure optimal usage of the user's viewing history for personalizing search ranking. An end-to-end model has the ability to learn members preferences directly from raw event data.

While this end-to-end approach is much more powerful from a modeling perspective, it introduces several challenges from a training and serving point of view. We now need to train the model on a much larger scale given the higher number of parameters and so that we can learn individual users' preferences accurately. On the serving side, the higher complexity of the model poses a latency challenge and requires non-trivial serving optimization.

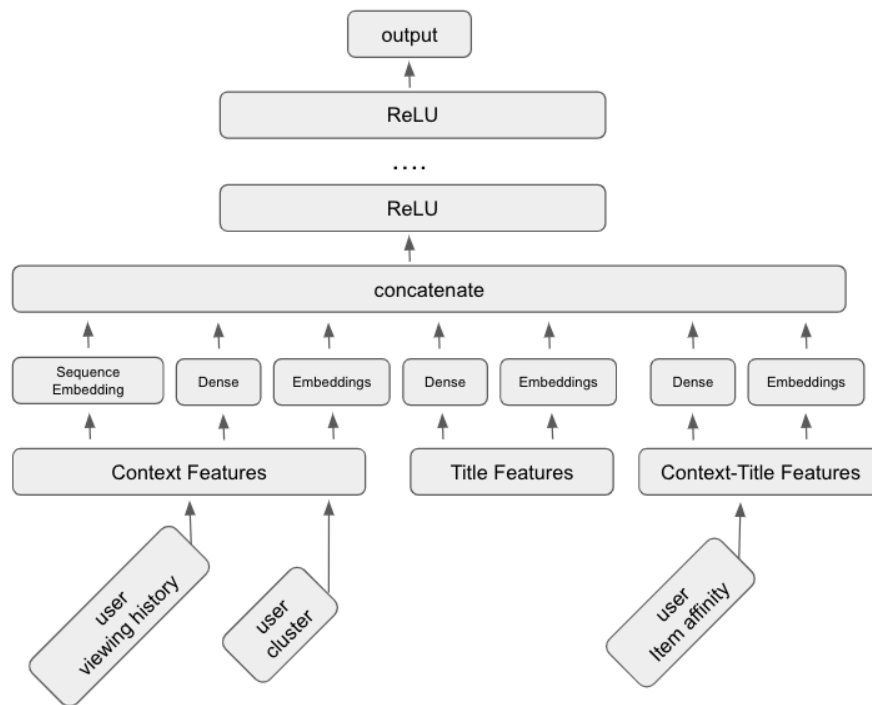


Figure 1: High level overview of the Neural Network ranker

4 CONCLUSION AND FUTURE WORK

In this paper, we described the Netflix Instant Search personalization problem and discussed some modeling choices and trade-offs of different approaches.

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REFERENCES

- [1] Dhruv Arya, Viet Ha-Thuc, and Shakti Sinha. 2015. Personalized Federated Search at LinkedIn. In *Proceedings of the 24th ACM International Conference on Information and Knowledge Management* (Melbourne, Australia) (CIKM '15). Association for Computing Machinery, New York, NY, USA, 1699–1702. <https://doi.org/10.1145/2806416.2806615>
- [2] Paul N. Bennett, Ryan W. White, Wei Chu, Susan T. Dumais, Peter Bailey, Fedor Borisov, and Xiaoyuan Cui. 2012. Modeling the Impact of Short- and Long-Term Behavior on Search Personalization. In *Proceedings of the 35th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Portland, Oregon, USA) (SIGIR '12). Association for Computing Machinery, New York, NY, USA, 185–194. <https://doi.org/10.1145/2348283.2348312>
- [3] Mark J. Carman, Fabio Crestani, Morgan Harvey, and Mark Baillie. 2010. Towards Query Log Based Personalization Using Topic Models. In *Proceedings of the 19th ACM International Conference on Information and Knowledge Management* (Toronto, ON, Canada) (CIKM '10). Association for Computing Machinery, New York, NY, USA, 1849–1852. <https://doi.org/10.1145/1871437.1871745>
- [4] Mihajlo Grbovic. 2017. Search Ranking And Personalization at Airbnb. In *Proceedings of the Eleventh ACM Conference on Recommender Systems* (Como, Italy) (RecSys '17). Association for Computing Machinery, New York, NY, USA, 339–340. <https://doi.org/10.1145/3109859.3109920>
- [5] Viet Ha-Thuc and Shakti Sinha. 2016. Learning to Rank Personalized Search Results in Professional Networks. In *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Pisa, Italy) (SIGIR '16). Association for Computing Machinery, New York, NY, USA, 461–462. <https://doi.org/10.1145/2911451.2927018>
- [6] Sudarshan Lamkhede and Sudeep Das. 2019. Challenges in search on streaming services: netflix case study. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1371–1374.
- [7] Jaime Teevan, Susan T. Dumais, and Eric Horvitz. 2005. Personalizing Search via Automated Analysis of Interests and Activities. In *Proceedings of the 28th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval* (Salvador, Brazil) (SIGIR '05). Association for Computing Machinery, New York, NY, USA, 449–456. <https://doi.org/10.1145/1076034.1076111>